

BOUNDARY RIGIDITY AND STABLE LEFT–RIGHT MODULI AT GENERIC DEGREE THREE

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ABSTRACT. We exhibit a one-parameter family of polynomial Keller maps $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$, all of generic degree three, whose members remain pairwise inequivalent after adjoining arbitrarily many identity variables and applying arbitrary polynomial changes of source and target coordinates. The family is obtained from the quadratic cubic-frame data

$$A_{\alpha}(c) = c + \alpha c^2, \quad B_{\alpha,\beta}(c) = -2 - 4\alpha c + \beta c^2.$$

For $\alpha \neq 0$, the complete stable left–right invariant is

$$q = \beta/\alpha^2.$$

The line $\alpha = 0$ is a single ordinary left–right orbit, stably distinct from every $\alpha \neq 0$ orbit.

The invariant is recovered from the nonproperness divisor. After normalizing its singular discriminant component, one obtains an affine plane with two intrinsically marked curves. A unique-factorization argument shows that every automorphism of every affine cylinder preserving this marked pair fixes q . A separate singular-incidence argument treats the exceptional value $q = -2$. Exact symbolic calculations verify the discriminant, normalization, conductor, exceptional fiber, and the explicit gauge transformation on the line $\alpha = 0$.

More generally, on the squarefree coprime cubic-frame locus, the complete stable invariant is the deleted infinity scheme $V(A/c)$ decorated by the transverse derivative $B|_{V(A/c)}$. Stable and ordinary left–right equivalence coincide there. This yields stable-moduli spaces of arbitrarily large dimension, all at generic degree three and, for each family, fixed ordinary degree.

Working proof draft. The exact symbolic verifier has been rerun from the recovered source artifact. Independent implementation and specialist review remain submission requirements.

1. INTRODUCTION

The recent counterexample to the Jacobian conjecture gives a polynomial map $\mathbb{A}_{\mathbb{C}}^3 \rightarrow \mathbb{A}_{\mathbb{C}}^3$ with nonzero constant Jacobian determinant that is not an automorphism [1, 6]. We credit Akhil Mathew as the primary human source of the problem: Mathew prompted Levent Alpöge, Alpöge prompted Fable, Fable produced the example, and Alpöge announced it publicly [7]. The construction was quickly placed in larger families. In particular, public

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accounts allow arbitrary one-variable deformation data and produce counterexamples of every generic degree [7, 3]. Varying generic degree already gives inequivalent maps.

This paper asks a narrower question:

Can a family of counterexamples of one fixed generic degree carry a modulus that survives arbitrary polynomial source and target changes, even after stabilization?

The answer is yes already at generic degree three.

Let

$$F, H: \mathbb{A}^n \longrightarrow \mathbb{A}^n$$

be polynomial maps. They are *polynomially left–right equivalent* if there are polynomial automorphisms Φ, Ψ of $\mathbb{A}^n$ such that

$$H \circ \Phi = \Psi \circ F.$$

They are *stably polynomially left–right equivalent* if, for some $m \geq 0$, the maps

$$F \times \text{id}_{\mathbb{A}^m}, \quad H \times \text{id}_{\mathbb{A}^m}$$

are polynomially left–right equivalent. We use this definition throughout; “stable equivalence” has other meanings in the Jacobian-conjecture literature.

For $\alpha, \beta \in \mathbb{C}$, consider

$$A_\alpha(C) = C + \alpha C^2, \quad B_{\alpha, \beta}(C) = -2 - 4\alpha C + \beta C^2. \quad (1.1)$$

These polynomials define a Keller map $G_{\alpha, \beta}$ by the cubic-frame construction in section 2. Every member has generic degree three and is nonproper. Our main theorem classifies the entire two-parameter family.

Theorem 1.1 (Complete stable classification). *The stable polynomial left–right orbit set of the family $\{G_{\alpha, \beta}\}$ is*

$$\{\mathcal{O}_0\} \sqcup \{\mathcal{O}_q : q \in \mathbb{C}\},$$

where

$$\mathcal{O}_0 = \{G_{0, \beta} : \beta \in \mathbb{C}\}$$

and

$$\mathcal{O}_q = \{G_{\alpha, \beta} : \alpha \neq 0, \beta/\alpha^2 = q\}.$$

More explicitly:

- (i) all maps $G_{0, \beta}$ are ordinarily left–right equivalent;
- (ii) no $G_{0, \beta}$ is stably equivalent to a member with $\alpha \neq 0$; and
- (iii) if $\alpha, \alpha' \neq 0$, then

$$G_{\alpha, \beta} \sim_{\text{stable}} G_{\alpha', \beta'} \iff \frac{\beta}{\alpha^2} = \frac{\beta'}{\alpha'^2}.$$

The weighted-scaling invariance of β/α^2 is immediate. The content of theorem 1.1 is completeness under *arbitrary* polynomial left–right equivalence and after arbitrary affine stabilization.

Our invariant comes from the set at which the map is not proper. For a dominant equidimensional polynomial map this set, if nonempty, is a hypersurface by Jelonek’s theorem [4]. In our normalized one-parameter slice it has a singular discriminant component D_q and a plane component P . The normalization of D_q is an affine plane containing two intrinsic marked curves:

$$L_q : 3c(c+1)t + qc^2 - 4c - 2 = 0, \quad M : c+1 = 0.$$

The pair (L_q, M) remembers q , even after taking a product with affine space. This gives the necessity in theorem 1.1.

The existence of left–right moduli for nonproper étale maps on other affine varieties has important precedent: Dubouloz and Palka constructed high-dimensional families on affine pseudo-planes [2]. Our theorem concerns polynomial self-maps of affine space arising directly from the new three-dimensional construction. The general marked-root framework and its unstabilized equivalence criterion are developed in the companion paper [5]. The proof below includes all of the special geometry needed for stable separation in the quadratic slice.

2. THE QUADRATIC CUBIC-FRAME FAMILY

Put

$$c = 2x - 3x^2y - x^3z, \quad t = y + \frac{1}{x}, \quad r = \frac{2}{x}. \quad (2.1)$$

For $A = A_\alpha$ and $B = B_{\alpha,\beta}$, let

$$h(T, C) = A(C)T^3 + B(C)T^2$$

and define

$$b = r - \frac{\partial h}{\partial T}(t, c), \quad 2a = h(t, c) + tb. \quad (2.2)$$

Although (2.1) uses $1/x$, the two expressions in (2.2) are polynomials. We write

$$G_{\alpha,\beta} = (a, b, c) : \mathbb{A}_{x,y,z}^3 \longrightarrow \mathbb{A}_{a,b,c}^3.$$

Proposition 2.1. *For every $\alpha, \beta \in \mathbb{C}$, the map $G_{\alpha,\beta}$ is polynomial,*

$$\det DG_{\alpha,\beta} = -2,$$

and its generic degree is three.

Proof. Set

$$\xi = 1 + xy, \quad d = 2 - 3xy - x^2z,$$

so that $t = \xi/x$ and $c = xd$. Clearing denominators in b gives

$$b = \frac{2 + 4\xi - 3d\xi^2}{x} - 3\alpha d^2\xi^2 + 8\alpha d\xi - 2\beta xd^2\xi.$$

The numerator of the displayed fraction vanishes at $x = 0$, hence is divisible by x . Similarly,

$$2a = \frac{2\xi - 2d\xi^3 + 2\xi^2}{x^2} + \frac{4\alpha d\xi^2 - 2\alpha d^2\xi^3}{x} - \beta d^2\xi^2.$$

The first numerator is divisible by x^2 , and the second by x ; this follows directly on substituting $\xi = 1 + xy$ and $d = 2 - 3xy - x^2z$. Thus a and b are polynomial.

On $x \neq 0$, the coordinate changes in (2.1) give

$$\det \frac{\partial(a, b, c)}{\partial(t, r, c)} = \frac{r}{2}, \quad \det \frac{\partial(t, r, c)}{\partial(x, y, z)} = -2x.$$

Since $r = 2/x$, their product is -2 . Polynomiality extends this identity across $x = 0$.

For a target point (a, b, c) , introduce

$$P_{a,b,c}(T) = A(c)T^3 + B(c)T^2 + bT - 2a. \quad (2.3)$$

The construction gives

$$P_{G_{\alpha,\beta}(x,y,z)}(t) = 0, \quad P'_{G_{\alpha,\beta}(x,y,z)}(t) = \frac{2}{x}.$$

Conversely, every finite simple root t reconstructs

$$x = \frac{2}{P'(t)}, \quad y = t - \frac{P'(t)}{2}, \quad z = \frac{2x - 3x^2y - c}{x^3}. \quad (2.4)$$

A general cubic (2.3) has three finite simple roots, giving three source points. $\square$

Remark 2.2. The determinant in proposition 2.1 is -2 , rather than 1, only because of our target normalization. A linear target rescaling gives the usual Keller normalization.

2.1. The ordinary scaling orbit. Suppose $\alpha, \alpha' \neq 0$, and put

$$u = \frac{\alpha}{\alpha'}.$$

If

$$\frac{\beta}{\alpha^2} = \frac{\beta'}{\alpha'^2},$$

then direct substitution gives

$$G_{\alpha',\beta'}\left(ux, \frac{y}{u}, \frac{z}{u^2}\right) = \left(\frac{a}{u^2}, \frac{b}{u}, uc\right) \quad \text{when} \quad (a, b, c) = G_{\alpha,\beta}(x, y, z). \quad (2.5)$$

Thus equality of $q = \beta/\alpha^2$ is sufficient for ordinary equivalence. Every nonzero- α orbit therefore has the normalized representative

$$G_q := G_{1,q},$$

with

$$A(c) = c(c+1), \quad B_q(c) = qc^2 - 4c - 2. \quad (2.6)$$

3. THE INTRINSIC NONPROPERNESS DIVISOR

For a dominant polynomial map F , write S_F for its reduced nonproperness set. Polynomial source automorphisms leave S_F unchanged, target automorphisms carry it isomorphically to the new nonproperness set, and

$$S_{F \times \text{id}_{\mathbb{A}^m}} = S_F \times \mathbb{A}^m. \quad (3.1)$$

The last identity follows either from the valuative criterion or directly from sequences escaping to infinity.

The discriminant of the normalized fiber cubic

$$P_{a,b,c}(T) = A(c)T^3 + B_q(c)T^2 + bT - 2a$$

is

$$\Delta_q = B_q^2 b^2 - 4Ab^3 + 8B_q^3 a - 108A^2 a^2 - 36AB_q ab. \quad (3.2)$$

For $q \neq -2$, let

$$D_q = V(\Delta_q).$$

For $q = -2$, the polynomial Δ_{-2} has content $c + 1$; in that case put

$$D_{-2} = V\left(\frac{\Delta_{-2}}{c+1}\right).$$

In every case let

$$P = V(c+1).$$

Proposition 3.1. *For every $q \in \mathbb{C}$,*

$$S_{G_q} = D_q \cup P.$$

These are the two irreducible components. For $q \neq -2$, the component D_q is singular and P is smooth.

Proof. For $q \neq -2$, the polynomials A and B_q are coprime. Regarding Δ_q as a quadratic in a , one computes

$$\text{Disc}_a(\Delta_q) = 64(B_q^2 - 3Ab)^3.$$

The right side is not a square in $\mathbb{C}(b, c)$, so Gauss's lemma gives the irreducibility of Δ_q .

At a general point of D_q , two roots of the inverse cubic coalesce. Along a splitting family their derivative tends to zero, so (2.4) gives $x \rightarrow \infty$. Thus D_q lies in the nonproperness set. Along $c = -1$, the cubic loses its root at infinity; this is the unmarked zero of $A = c(c+1)$, and the reconstruction again escapes. Hence P is also a nonproperness component. Away from $D_q \cup P$, all roots are finite and simple and (2.4) is regular, proving the assertion for $q \neq -2$.

At $q = -2$, one has $B_{-2} = -2(c+1)^2$. Dividing the discriminant by its coefficient content $c+1$ gives the unique primitive nonplane component, while the same lost-root argument gives P . Irreducibility of the primitive discriminant follows by applying the preceding quadratic argument over $\mathbb{C}(c)$ and then Gauss's lemma. This is also the special case of the all-multiplicity nonproperness theorem in [5].

Finally, for $q \neq -2$ the normalization calculation in proposition 4.1 shows that D_q is singular, while P is a plane. $\square$

4. NORMALIZATION AND THE MARKED PAIR

Assume throughout this section that $q \neq -2$. A repeated root t of the inverse cubic gives

$$\begin{aligned} b &= -3A(c)t^2 - 2B_q(c)t, \\ a &= -A(c)t^3 - \frac{1}{2}B_q(c)t^2. \end{aligned} \tag{4.1}$$

Let

$$\nu_q : \mathbb{A}_{c,t}^2 \longrightarrow D_q$$

be the map (4.1), and put

$$H_q(c, t) = 3c(c+1)t + qc^2 - 4c - 2. \tag{4.2}$$

Proposition 4.1. *The map ν_q is the normalization of D_q . Moreover,*

$$\nu_q^{-1}(\text{Sing } D_q) = L_q := V(H_q).$$

Proof. On the parametrization, t satisfies

$$3At^2 + 2B_qt + b = 0, \tag{4.3}$$

$$B_qt^2 + 2bt - 6a = 0. \tag{4.4}$$

Because $\gcd(A, B_q) = 1$, choose $\rho, \sigma \in \mathbb{C}[c]$ with $\rho A + \sigma B_q = 1$. The sum of $\rho/3$ times (4.3) and σ times (4.4) is a monic quadratic equation for t over $\mathbb{C}[D_q]$. Hence ν_q is finite.

Direct substitution gives

$$B_q^2 - 3Ab = H_q^2, \quad -18Aa - B_qb = 2tH_q^2. \tag{4.5}$$

Thus t is rationally recovered away from $H_q = 0$, so ν_q is birational. Since $\mathbb{A}^2$ is normal, it is the normalization.

The pullbacks of the three partial derivatives of Δ_q are

$$(\Delta_q)_a \circ \nu_q = 8H_q^3, \tag{4.6}$$

$$(\Delta_q)_b \circ \nu_q = -4tH_q^3, \tag{4.7}$$

$$(\Delta_q)_c \circ \nu_q = -4t^2(2cq + 2ct + t - 4)H_q^3. \tag{4.8}$$

The first equality shows that all three vanish simultaneously exactly on $H_q = 0$. $\square$

The plane component P marks a second curve upstairs. Write

$$d = q + 2.$$

On $c = -1$, formulas (4.1) become

$$a = -\frac{d}{2}t^2, \quad b = -2dt. \tag{4.9}$$

Since $d \neq 0$, the inverse image of $D_q \cap P$ is the line

$$M := V(c+1) \subset \mathbb{A}_{c,t}^2,$$

and $M \rightarrow D_q \cap P$ is an isomorphism. Notice also that

$$H_q|_M = q+2 \neq 0,$$

so $L_q \cap M = \emptyset$.

The intrinsic data recovered from S_{G_q} are therefore

$$(\mathbb{A}_{c,t}^2, L_q, M). \quad (4.10)$$

The next lemma is the heart of the stable argument.

Lemma 4.2 (Rigidity of the marked cylinder). *Let $q, q' \neq -2$ and $m \geq 0$. Suppose a polynomial automorphism of $\mathbb{A}^{2+m}$ carries*

$$L_q \times \mathbb{A}^m \text{ to } L_{q'} \times \mathbb{A}^m \quad \text{and} \quad M \times \mathbb{A}^m \text{ to } M \times \mathbb{A}^m.$$

Then $q = q'$.

Proof. Let C, T be the pullbacks of the first two coordinates on the target cylinder. Since the prime principal divisor $M \times \mathbb{A}^m$ is preserved, unique factorization gives

$$C+1 = \lambda(c+1) \quad (4.11)$$

for some $\lambda \in \mathbb{C}^*$. For $q \neq -2$, the polynomial H_q is primitive and linear in t , hence irreducible. Preservation of the other prime divisor similarly gives

$$H_{q'}(C, T) = \kappa H_q(c, t) \quad (4.12)$$

for some $\kappa \in \mathbb{C}^*$.

The right side of (4.12) is independent of the stabilization variables and linear in t . Since C is the nonconstant polynomial in (4.11), it follows that

$$T = \mu(c)t + h(c)$$

for polynomials $\mu, h \in \mathbb{C}[c]$. Comparing the coefficients of t in (4.12) and cancelling $c+1$ gives

$$\lambda(\lambda(c+1)-1)\mu(c) = \kappa c. \quad (4.13)$$

The linear polynomial $\lambda(c+1)-1$ must divide c . Therefore $\lambda = 1$, and (4.13) gives $\mu = \kappa$.

The constant term of (4.12) is now

$$3c(c+1)h(c) + B_{q'}(c) = \kappa B_q(c). \quad (4.14)$$

At $c = 0$, both B_q and $B_{q'}$ equal -2 , so $\kappa = 1$. At $c = -1$, equation (4.14) gives $q' + 2 = q + 2$. $\square$

Theorem 4.3. *If $q, q' \neq -2$, then G_q and $G_{q'}$ are stably polynomially left-right equivalent if and only if $q = q'$.*

Proof. Sufficiency is immediate. For necessity, suppose the stabilizations by $\mathbb{A}^m$ are left-right equivalent. By (3.1), the target automorphism identifies

$$(D_q \cup P) \times \mathbb{A}^m \quad \text{with} \quad (D_{q'} \cup P) \times \mathbb{A}^m.$$

The plane component is smooth and the discriminant component is singular, so the two components cannot be exchanged. We obtain an isomorphism

$$D_q \times \mathbb{A}^m \simeq D_{q'} \times \mathbb{A}^m$$

that preserves the intersection with the plane component.

Normalization commutes with adjoining polynomial variables. The isomorphism therefore lifts uniquely to an automorphism of $\mathbb{A}^2 \times \mathbb{A}^m$. The inverse image of the singular locus is $L_q \times \mathbb{A}^m$, and the inverse image of the plane intersection is $M \times \mathbb{A}^m$. The lifted automorphism preserves this marked pair, so lemma 4.2 gives $q = q'$. $\square$

5. THE EXCEPTIONAL VALUE

At $q = -2$,

$$B_{-2}(c) = -2(c+1)^2$$

and Δ_{-2} acquires the factor $c+1$. Let

$$\Delta_{-2}^{\text{res}} = \frac{\Delta_{-2}}{c+1}, \quad D_{-2} = V(\Delta_{-2}^{\text{res}}).$$

Proposition 5.1. *The map G_{-2} is not stably left-right equivalent to G_q for any $q \neq -2$.*

Proof. On the plane $c = -1$, exact substitution gives

$$\Delta_{-2}^{\text{res}}|_{c=-1} = 4b^3.$$

Thus the reduced intersection $D_{-2} \cap P$ is the line $b = 0$. At the point

$$p = (a, b, c) = (0, 0, -1),$$

all three first partial derivatives of Δ_{-2}^{res} vanish. Hence

$$\text{Sing}(D_{-2}) \cap P \neq \emptyset.$$

For $q \neq -2$, by contrast,

$$\left. \frac{\partial \Delta_q}{\partial a} \right|_{c=-1} = 8(q+2)^3 \neq 0.$$

Consequently D_q is smooth along its entire intersection with P , and

$$\text{Sing}(D_q) \cap P = \emptyset.$$

Whether the singular locus of the unique nonplane component meets the plane component is intrinsic to the reduced nonproperness divisor. The same is true after taking a product with affine space. Equation (3.1) therefore separates G_{-2} from every other G_q stably. $\square$

Corollary 5.2. *For all $q, q' \in \mathbb{C}$,*

$$G_q \sim_{\text{stable}} G_{q'} \iff q = q'.$$

Proof. Combine theorem 4.3 and proposition 5.1. $\square$

6. THE LINE $\alpha = 0$ IS GAUGE

It remains to understand the parameter line not covered by the normalization $\alpha = 1$. When $\alpha = 0$,

$$A(C) = C, \quad B(C) = -2 + \beta C^2.$$

Put

$$s = \frac{\beta}{3}, \quad c(x, y, z) = 2x - 3x^2y - x^3z,$$

and define

$$\Theta_s(x, y, z) = \left(x, y + sc, z - 3s\frac{c}{x} \right). \quad (6.1)$$

This is polynomial because

$$\frac{c}{x} = 2 - 3xy - x^2z.$$

It preserves c , has Jacobian determinant one, and has inverse Θ_{-s} .

Set

$$\ell_s(c) = 3s^2c^3 - 4sc, \quad \delta_s(c) = s^3c^4 - 2s^2c^2, \quad (6.2)$$

and define the triangular target automorphism

$$\Xi_s(a, b, c) = \left(a - \frac{1}{2}\delta_s(c) - \frac{sc}{2}b, b + \ell_s(c), c \right). \quad (6.3)$$

Proposition 6.1. *For every $\beta \in \mathbb{C}$,*

$$G_{0,\beta} = \Xi_{\beta/3} \circ G_{0,0} \circ \Theta_{\beta/3}.$$

In particular, the entire line $\alpha = 0$ is one ordinary left-right orbit.

Proof. Because Θ_s fixes c , it sends the framed root $t = y + 1/x$ to $t + sc$. Substitute this translation in

$$cT^3 - 2T^2$$

and compare the coefficients of T with

$$cT^3 + (-2 + 3sc^2)T^2.$$

The residual linear and constant terms are precisely the target shears ℓ_s and δ_s in (6.2). Substitution in (2.2) gives the displayed identity. All expressions are polynomial, and Θ_s, Ξ_s are triangularly invertible. $\square$

Proposition 6.2. *No member with $\alpha = 0$ is stably left-right equivalent to a member with $\alpha \neq 0$.*

Proof. For $\alpha = 0$, the polynomial $A(C) = C$ has only the marked root. The reduced nonproperness set therefore consists solely of its irreducible primitive discriminant component. For $\alpha \neq 0$, the second root $-1/\alpha$ contributes a plane component in addition to the primitive discriminant component, as in proposition 3.1. Thus the two nonproperness sets have respectively one and two irreducible components. Products with $\mathbb{A}^m$ preserve this component count, and (3.1) makes it a stable left-right invariant. $\square$

Proof of theorem 1.1. The first assertion is proposition 6.1; the second is proposition 6.2. For nonzero α, α' , equality of β/α^2 gives the ordinary equivalence (2.5). Conversely, normalize both maps to $\alpha = 1$ using the same formula and apply corollary 5.2. $\square$

7. EXACT VERIFICATION AND PROOF BOUNDARY

The accompanying verifier uses exact SymPy arithmetic. It checks:

- (1) the discriminant (3.2) and its discriminant as a quadratic in a ;
- (2) the normalization formulas (4.1);
- (3) both rational-recovery identities (4.5);
- (4) all three gradient factorizations (4.6)–(4.8);
- (5) the marked-line formulas (4.9);
- (6) the factorization and singular-incidence calculation at $q = -2$;
- (7) the coefficient identity used in lemma 4.2;
- (8) the identity that gauges away β when $\alpha = 0$; and
- (9) the Jacobian identity $\det D\Theta_s = 1$.

The unchanged recovered verifier was rerun in a fresh environment with SymPy declared explicitly. It printed

ALL q-ORBIT CHECKS PASSED,

and regenerated a JSON audit byte-for-byte identical to the recovered reference output.

These computations verify the displayed polynomial identities. They do not replace the geometric arguments that the nonproperness divisor is intrinsic, that normalization lifts an isomorphism, or that the two marked curves are recovered as asserted. Those are conventional proofs in sections 3 and 4. A genuinely independent CAS implementation and expert review of the stable-cylinder argument remain release gates.

8. CONSEQUENCES AND QUESTIONS

Theorem 1.1 gives an affine line of stable left–right orbits at one fixed generic degree, together with one additional degenerate orbit. The modulus is not imposed as part of a framing: it is reconstructed from the intrinsic boundary geometry of the map.

The proof deliberately leaves broader claims outside its scope. In particular, it does not classify all cubic-frame counterexamples, all generic-degree-three Keller maps, or the full stable moduli functor.

Question 8.1. *Which larger finite-dimensional cubic-frame families admit complete stable invariants recovered from their normalized nonproperness divisors?*

Question 8.2. *Can the marked-cylinder rigidity argument be formulated as a general cancellation theorem for pairs $(\mathbb{A}^2, L, M)$?*

Question 8.3. *What scheme or stack represents deformations of the maps in theorem 1.1 modulo stable polynomial left–right equivalence?*

Question 8.4 (Compactification across boundary collisions). *Is there a natural compactification of this stable-moduli problem that retains nonreduced infinity schemes, conductor or jet data, and the exceptional parameter values where boundary components collide? Such an object should make the exceptional strata geometric rather than a list of ad hoc cases.*

APPENDIX A. THE RECIPROCAL CUBIC FAMILY

The family in the body gives a clean affine line of stable orbits. A second construction gives much larger fixed-generic-degree families and boundary curves of arbitrary genus. We record it separately because the algebraic construction is exact, while the last stable-separation step still requires a specialist audit of the birational argument.

Put

$$\begin{aligned} A &= 1 + xy, & B &= A^2z + y^2(4 + 3xy), & P &= AB, \\ Q &= y + 3xB, & R &= 2x - 3x^2y - x^3z. \end{aligned}$$

Thus (P, Q, R) is the unnormalized degree-seven map. On $A \neq 0$, set

$$s = \frac{x}{A}.$$

Then $Ps = xB$ and $1 - ys = A^{-1}$.

A.1. Pole cancellation. Let $g(p) \in \mathbb{C}[p]$. In the rational coordinates (P, s, y) , consider

$$q = y + 3Pg(P)s, \quad r = 2Pg(P)s^3 - qs^2 + 2s.$$

Proposition A.1 (Reciprocal pole-cancellation criterion). *The prescription (P, q, r) extends to a polynomial self-map of $\mathbb{A}^3$ if and only if*

$$g(0) = 1, \quad g'(0) = 0.$$

Equivalently, $g(p) = 1 + p^2\eta(p)$ for a unique $\eta \in \mathbb{C}[p]$.

Proof. The second coordinate is $q = y + 3xBg(P)$. Comparing the third coordinate with R gives

$$r - R = -\frac{x^3B}{A^2}(g(P) - 1).$$

Write $g(P) - 1 = c_0 + c_1P + P^2\eta(P)$. Since $\gcd(A, x^3B) = 1$ in $\mathbb{C}[x, y, z]$, polynomiality forces $A^2 \mid c_0 + c_1AB$. Reducing first modulo A , and then after one division by A , gives $c_0 = c_1 = 0$. The converse follows from

$$r - R = -x^3B^3\eta(P).$$

□

For $\eta \in \mathbb{C}[p]$, define

$$\begin{aligned} Q_\eta &= Q + 3xA^2B^3\eta(P), \\ R_\eta &= R - x^3B^3\eta(P), \\ \mathcal{G}_\eta &= (P, Q_\eta, R_\eta), \end{aligned} \tag{A.1}$$

and write $g(p) = 1 + p^2\eta(p)$.

Theorem A.2 (Algebraic properties of the reciprocal family). *For every η ,*

$$\det J\mathcal{G}_\eta = -2.$$

The map has generic degree three and generic Galois group S_3 .

Proof. In rational coordinates,

$$Q_\eta = y + 3Pg(P)s, \quad R_\eta = 2Pg(P)s^3 - Q_\eta s^2 + 2s.$$

Direct differentiation gives

$$\det \frac{\partial(P, Q_\eta, R_\eta)}{\partial(P, s, y)} = -\frac{2}{A}, \quad \det \frac{\partial(P, s, y)}{\partial(x, y, z)} = A.$$

Their product is -2 .

For target coordinates (p, q, r) , the primitive element satisfies

$$\Omega_{p,q,r}(S) = 2pg(p)S^3 - qS^2 + 2S - r = 0. \tag{A.2}$$

Moreover $y = q - 3pg(p)s$, so $\mathbb{C}(x, y, z) = \mathbb{C}(p, q, s)$. Over $\mathbb{C}(p, q)$, the right side defining r is a degree-three rational function of s , proving generic degree three and irreducibility. The discriminant is $4\Delta_\eta$, where

$$\Delta_\eta = q^2 - 16pg - q^3r + 18pgqr - 27p^2g^2r^2.$$

As a quadratic in r , its discriminant is $(q^2 - 12pg)^3$, which is not a square in the generic function field. The irreducible cubic therefore has Galois group S_3 . $\square$

A.2. The image and finite completion. Define

$$\Gamma_\eta : \quad q^2 = 12pg(p), \quad 3qr = 4,$$

and

$$\Lambda_\eta : \quad g(p) = 0, \quad qr = 1.$$

Theorem A.3 (Exact image). *Set-theoretically over $\mathbb{C}$,*

$$\mathbb{A}^3 \setminus \mathcal{G}_\eta(\mathbb{A}^3) = \Gamma_\eta \sqcup \Lambda_\eta.$$

Proof. A simple root s of (A.2) reconstructs a unique source point. Indeed, with

$$D = \Omega'(s), \quad a = \frac{2}{D}, \quad x = as, \quad y = q - 3pg(p)s, \quad b = \frac{p}{a},$$

put

$$z = \frac{b - y^2(4 + 3xy)}{a^2}.$$

Then $1 + xy = a$, $B = b$, and substitution gives the target (p, q, r) .

For $p \neq 0$, a target is therefore omitted precisely when Ω has no simple root. If $g(p) \neq 0$, this happens exactly when the cubic is a cube of a linear polynomial, which gives Γ_η . If $g(p) = 0$, the equation is quadratic and has no simple root exactly when $qr = 1$, which gives Λ_η .

The plane $p = 0$ is in the image. If $p = q = 0$, take $(x, y, z) = (r/2, 0, 0)$. If $p = 0$ and $q \neq 0$, take

$$x = \frac{2}{q}, \quad y = -\frac{q}{2}, \quad z = \frac{5q^2}{4} - \frac{\eta(0)q^6}{64} - \frac{rq^3}{8}.$$

Neither displayed omitted locus meets this plane. $\square$

Let

$$\mathcal{X}_\eta = V(2pg(p)u^3 - qu^2v + 2uv^2 - rv^3) \subset \mathbb{A}_{p,q,r}^3 \times \mathbb{P}_{[u:v]}^1.$$

The projection $\mathcal{X}_\eta \rightarrow \mathbb{A}^3$ is finite flat of rank three. The source embeds by

$$(x, y, z) \mapsto (\mathcal{G}_\eta(x, y, z), [x : A]).$$

If $\mathcal{R}_\eta$ is the repeated-root divisor and $\mathcal{I}_\eta = V(v, g(p))$, the reconstruction above in the two projective charts gives

$$\mathbb{A}^3 \simeq \mathcal{X}_\eta \setminus (\mathcal{R}_\eta \cup \mathcal{I}_\eta)$$

and hence

$$S_{\mathcal{G}_\eta}^{\text{red}} = V(g(p)\Delta_\eta(p, q, r)). \quad (\text{A.3})$$

This finite completion is also the natural frame for the moduli assertions below.

A.3. Curves and stable separation. The component Γ_η is a dense open of the normalization of

$$C_\eta : \quad u^2 = 12p(1 + p^2\eta(p)).$$

If $\eta \neq 0$ has degree d and the polynomial on the right is squarefree, then

$$g(C_\eta) = \left\lfloor \frac{d+2}{2} \right\rfloor.$$

Each distinct root of g contributes a rational $\mathbb{G}_m$ -component to Λ_η .

Theorem A.4 (Stable curve invariant—proof-draft form). *Assume that C_η and C_θ are smooth projective curves of positive genus. If $\mathcal{G}_\eta$ and $\mathcal{G}_\theta$ are polynomially left-right equivalent after stabilization, then*

$$C_\eta \simeq C_\theta.$$

Consequently the reciprocal family contains positive-dimensional families of fixed generic degree. Their boundary curves realize every positive genus.

Current proof draft. Stable left–right equivalence identifies the reduced non-properness loci after taking a product with affine space. In (A.3), the components arising from Λ_η are rational, whereas Γ_η has projective normalization C_η . The intended intrinsic argument recovers the nonrational curve from the maximal rationally connected quotient of a projective model of the normalized boundary cylinder. Affine stabilization adds only a rational factor, so this quotient is unchanged. It follows that the positive-genus curve associated with η must be isomorphic to the one associated with θ .

For each genus, take

$$g_{\tau,m}(p) = (1 - p^2)(1 - p^2/\tau^2) \prod_{j=3}^m (1 - p^2/\alpha_j^2).$$

Varying the symmetric branch set gives nonisomorphic curves for general τ . Moreover, $\eta = (g-1)/p^2$ is polynomial and the branch cross-ratios vary. $\square$

Remark A.5 (Required review before submission). The MRC-based recovery argument has not yet received an independent specialist audit. The exact map, Jacobian, cubic, image, completion, and genus formulas are not conditional; the theorem-level stable-separation conclusion should remain labeled “proof draft” until that audit is complete.

A.4. The collision-preserving line. For

$$\eta_\lambda(p) = \frac{\lambda}{3}(4p+1), \quad g_\lambda(p) = 1 + \frac{\lambda}{3}p^2(4p+1),$$

the projective normalization is the short Weierstrass curve with

$$j(\lambda) = \frac{1728\lambda}{\lambda + 324}.$$

For $\lambda \neq 0, -324$, the omitted set has one elliptic component and three rational components. At $\lambda = 0$, it has one component $\mathbb{G}_m$. At $\lambda = -324$,

$$g_{-324}(p) = -(6p+1)^2(12p-1),$$

and the reduced omitted locus has three components: two copies of $\mathbb{G}_m$ and a rational curve whose normalization is $\mathbb{P}^1$ minus six points. Subject to the stable-recovery step in theorem A.4, these data give a complete pairwise separation of the λ -line.

A.5. Degree and framed moduli. If

$$g(p) = 1 + c_2p^2 + \cdots + c_mp^m, \quad c_m \neq 0,$$

then

$$\deg \mathcal{G}_\eta = 7m + 6.$$

Thus degree 20 is the first nonbase member of this reciprocal normal form, while degree 27 is the first positive-dimensional quotient. These are normal-form statements, not absolute lower bounds under arbitrary polynomial equivalence. The separate coarse argument presently gives only

$$4 \leq d_{\text{positive genus}} \leq 20.$$

Fix a degree bound m for η and retain the finite completion, infinity section, and target coordinates as part of the framing. Let

$$V_m = \{\eta \in \mathbb{C}[p] : \deg \eta \leq m\}.$$

The residual source scaling acts on the coefficient of p^i with weight $-(2i+4)$. Exact pole cancellation and a calculation of frame-preserving source–target transformations give the framed moduli groupoid

$$[V_m/\mathbb{G}_m].$$

Its tangent complex at η is the standard two-term complex

$$\mathrm{Lie}(\mathbb{G}_m) \longrightarrow T_\eta V_m.$$

With the boundary marked, differentiation of (A.2) gives the corresponding logarithmic deformation complex. This is a statement about the framed reciprocal family; it is not a presentation of the full unframed Keller moduli stack.

APPENDIX B. ADDITIONAL RESTRICTED MODULI RESULTS

The results in this appendix concern narrower equivalence problems than the stable left–right problem solved in the body. They are retained because they identify useful intermediate moduli spaces, but they should not be read as classifications of all Keller maps of the stated degree.

B.1. A grading-preserving degree-five modulus. For the mixed grading

$$\deg(x, y, z) = (-1, 1, 2),$$

the equivariant Keller equation reduces to a finite coefficient system in the quotient variables $u = xy$ and $v = x^2z$. If

$$X = xf(u, v), \quad Y = x^{-1}g(u, v), \quad Z = x^{-2}h(u, v),$$

that system is

$$gJ(f, h) - 2hJ(f, g) + fJ(g, h) = 1.$$

Proposition B.1 (Restricted fixed-degree moduli). *Inside the analyzed generic-degree-five equivariant family, the grading-preserving left–right orbit space contains an affine one-parameter family. Its members are separated by the affine-scaling class of the critical-root configuration of the associated one-variable polynomial.*

Exact-elimination proof record. The weight restrictions leave finitely many coefficients. Solving the determinant equations gives a family whose remaining data include a one-variable critical polynomial. Every grading-preserving source and target transformation acts on its roots through the restricted affine scaling group. A cross-ratio-type invariant of the critical roots varies, which separates the displayed one-parameter subfamily. The archived elimination and transformation calculation is exact; the proposition makes no assertion about equivalences that do not preserve the grading. $\square$

B.2. The low-degree cubic frame and its based parameter. The pole-cancellation equations in the general cubic marked-root frame admit an exact normal form. In ordinary degree at most seven, every member of that frame is polynomially source–target equivalent to the base map. Thus the frame itself produces no additional unbased degree-seven modulus.

If the boundary frame and basepoint are fixed, a residual coefficient traditionally denoted κ remains. The exact restricted classification has two based classes:

$$\kappa = 0, \quad \kappa \neq 0.$$

The nonzero values are related by the residual scaling. On forgetting the frame, the literal coefficient can be moved by a determinant-one source–target translation; it is therefore not an invariant of ordinary left–right equivalence.

Coefficientwise formal trivializations of the nonzero based family exist by formal étaleness, but their spatial degrees grow with the deformation order. The recovered calculation gives componentwise degrees

$$(36, 34, 36), \quad (71, 69, 71)$$

for the first two recursively determined coefficients. These values are evidence of growth, not a lower-bound theorem for every possible trivialization.

Remark B.2 (Why these statements are separate). Proposition B.1 uses a grading as part of the equivalence problem. The κ -classification uses a fixed boundary frame and basepoint. Neither competes with the intrinsic stable invariant q proved in the body, nor with the reciprocal-family boundary invariant of appendix A.

APPENDIX C. BOUNDARY RESIDUES FOR GENERAL SQUAREFREE CUBIC FRAMES

The one-parameter invariant is the first member of a general construction. We record the squarefree statement here; the full standalone sequel source, including expanded calculations and examples, is retained under [supplements/](#).

Definition C.1 (Admissible frame). A pair $A, B \in \mathbb{C}[c]$ is admissible if

$$A(0) = 0, \quad A'(0) = 1, \quad B(0) = -2, \quad B'(0) = -2A''(0).$$

Using

$$c = 2x - 3x^2y - x^3z, \quad t = y + \frac{1}{x}, \quad r = \frac{2}{x},$$

define $G_{A,B} = (a, b, c)$ by

$$b = r - 3A(c)t^2 - 2B(c)t, \quad 2a = A(c)t^3 + B(c)t^2 + tb.$$

Proposition C.2 (Universal cubic frame). *For every admissible pair, $G_{A,B}$ is polynomial and*

$$\det DG_{A,B} = -2.$$

If $A \neq 0$, its generic degree is three.

Proof. The admissibility jets cancel the possible poles at $x = 0$. On $x \neq 0$, the change $(x, y, z) \mapsto (t, r, c)$ has determinant $-2x$, while $(t, r, c) \mapsto (a, b, c)$ has determinant $r/2$. Since $r = 2/x$, the product is -2 , and polynomiality extends it across $x = 0$. Finite simple roots of

$$A(c)T^3 + B(c)T^2 + bT - 2a$$

recover the source, giving generic degree three. $\square$

Assume now that A is squarefree and $\gcd(A, B) = 1$. Put

$$Z_A^\circ = V(A/c) \subset \mathbb{G}_m.$$

Homogenizing the inverse cubic gives

$$A(c)U^3 + B(c)U^2V + bUV^2 - 2aV^3 = 0.$$

At a root s of A , in the local coordinate V/U , the transverse derivative of the infinity section is $B(s)$. The root over $c = 0$ is retained in the affine source; the roots in Z_A° are deleted. Thus the natural decorated infinity datum is

$$(Z_A^\circ, \sigma_B = B|_{Z_A^\circ}).$$

The discriminant component is normalized by

$$\nu_{A,B}: \mathbb{A}_{c,t}^2 \longrightarrow D_{A,B},$$

$$b = -3A(c)t^2 - 2B(c)t, \quad a = -A(c)t^3 - \frac{1}{2}B(c)t^2.$$

The preimage of its singular locus is

$$L_{A,B} = V(3A(c)t + B(c)),$$

and the deleted infinity components lift to the vertical lines

$$M_s = V(c - s), \quad s \in Z_A^\circ.$$

Theorem C.3 (Stable boundary Torelli on the squarefree locus). *Let (A, B) and $(\tilde{A}, \tilde{B})$ be admissible squarefree coprime pairs. The following are equivalent:*

- (i) $G_{A,B}$ and $G_{\tilde{A},\tilde{B}}$ are stably polynomially left-right equivalent;
- (ii) they are ordinarily polynomially left-right equivalent;
- (iii) there is $u \in \mathbb{C}^*$ such that

$$\tilde{A}(uc) = uA(c), \quad \tilde{B}(uc) - B(c) \in cA(c)\mathbb{C}[c];$$

- (iv) multiplication by u identifies the decorated finite schemes

$$(Z_A^\circ, \sigma_B) \quad \text{and} \quad (Z_{\tilde{A}}^\circ, \sigma_{\tilde{B}}).$$

Proof. A stable equivalence preserves the reduced nonproperness divisor. Its unique singular nonplane component is the discriminant component, while its other components are the planes $c = s$, $s \in Z_A^0$. Passing to normalizations gives an automorphism of a cylinder over $\mathbb{A}_{c,t}^2$ preserving $L_{A,B}$ and the union of the vertical lines.

Write $Q_A = A/c$, and let C, T be the pullbacks of the normalization coordinates. Unique factorization gives

$$Q_{\tilde{A}}(C) = \lambda Q_A(c).$$

Hence C is algebraic over $\mathbb{C}(c)$. Since $\mathbb{C}(c)$ is algebraically closed in the purely transcendental cylinder function field, $C \in \mathbb{C}(c)$. Polynomiality of the automorphism and its inverse then force

$$C = uc + v.$$

The omitted vertical roots account for every root except the distinguished retained root, so the latter is preserved and $v = 0$. The normalization $A'(0) = \tilde{A}'(0) = 1$ yields

$$\tilde{A}(uc) = uA(c).$$

Preservation of the irreducible singular-preimage curve gives

$$3\tilde{A}(C)T + \tilde{B}(C) = \kappa(3A(c)t + B(c)).$$

The value at $c = 0$ gives $\kappa = 1$, and the admissibility jets imply

$$\tilde{B}(uc) - B(c) \in cA(c)\mathbb{C}[c].$$

This proves (i) $\Rightarrow$ (iii); (iii) and (iv) are equivalent by squarefreeness.

Conversely, after the diagonal change

$$(x, y, z) \mapsto (ux, y/u, z/u^2), \quad (a, b, c) \mapsto (a/u^2, b/u, uc),$$

condition (iii) reduces to

$$\tilde{B} - B = 3A\phi, \quad \phi \in c\mathbb{C}[c].$$

The source root translation

$$(x, y, z) \mapsto \left(x, y + \phi(c), z - 3\frac{\phi(c)}{x}\right)$$

is polynomial, preserves c , and shifts t by $\phi(c)$. Expanding the cubic supplies a triangular target automorphism, proving ordinary left-right equivalence. $\square$

Corollary C.4 (Arbitrarily large fixed-degree stable moduli). *If $\deg A = N + 1$, the generic squarefree orbit space is*

$$\frac{\{(s_i, v_i)_{i=1}^N \in (\mathbb{G}_m \times \mathbb{G}_m)^N : s_i \neq s_j\}}{\mathbb{G}_m \times S_N},$$

where $\mathbb{G}_m$ scales the s_i and fixes the v_i . Its generic dimension is $2N - 1$. Fixing a root configuration with no scaling symmetry leaves an N -dimensional

family of pairwise stably inequivalent maps. With the canonical interpolating representative for B , every map in such a family has

$$\begin{aligned}\det DG &= -2, & \mu(G) &= 3, \\ (\deg G_1, \deg G_2, \deg G_3) &= (4N + 7, 4N + 6, 4).\end{aligned}$$

Remark C.5 (The quadratic modulus). For

$$A = c + \alpha c^2, \quad B = -2 - 4\alpha c + \beta c^2,$$

the unique deleted infinity point is $-1/\alpha$, and

$$B(-1/\alpha) = 2 + \frac{\beta}{\alpha^2} = q + 2.$$

Thus q is the normalized transverse derivative at the deleted infinity section. Equivalently, it is the principal part of the rational Tschirnhaus translation needed to identify the cubic frames. At $q = -2$ that derivative vanishes and the deleted section becomes ramified.

Proof boundary. The exact verifier checks all displayed polynomial identities, the frame Jacobian, the marked infinity section, the root-translation gauge, and an explicit three-parameter interpolation family. The four load-bearing geometric steps—identification of the affine open in the finite completion, equality with the nonproperness locus, lifting stable equivalences through normalization, and rigidity of the marked vertical cylinder—still require independent specialist review.

AI AND COMPUTATIONAL DISCLOSURE

AI systems were used extensively in exploration, proof drafting, symbolic program development, source organization, and manuscript editing. They are not authors. The mathematical evidence for the computer-checked identities is the exact verifier and reproducible output described in section 7, not the output of a language model. Human specialist review has not yet been completed.

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